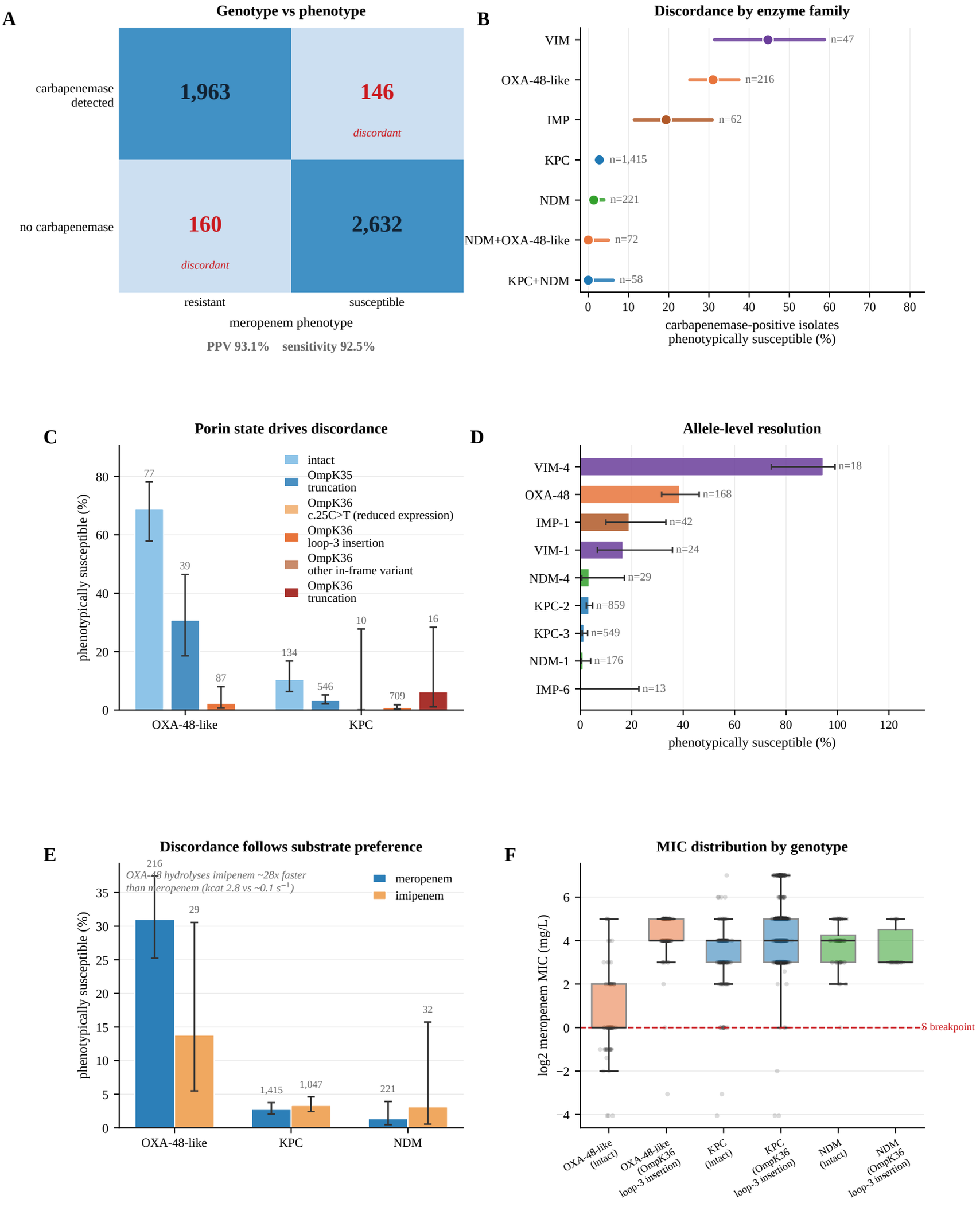

A
Genotype vs phenotype
carbapenemase detected
no carbapenemase
1,963
146
discordant
160
discordant
2,632
resistant
susceptible
meropenem phenotype
PPV 93.1%
sensitivity 92.5%
B
Discordance by enzyme family
VIM
OXA-48-like
IMP
KPC
NDM
NDM+OXA-48-like
KPC+NDM
n=47
n=216
n=62
n=1,415
n=221
n=72
n=58
0
10
20
30
40
50
60
70
80
carbapenemase-positive isolates phenotypically susceptible (%)
C
Porin state drives discordance
phenotypically susceptible (%)
intact
OmpK35 truncation
OmpK36 c.25C>T (reduced expression)
OmpK36 loop-3 insertion
OmpK36 other in-frame variant
OmpK36 truncation
77
39
87
134
546
10
709
16
OXA-48-like
KPC
D
Allele-level resolution
VIM-4
OXA-48
IMP-1
VIM-1
NDM-4
KPC-2
KPC-3
NDM-1
IMP-6
n=18
n=168
n=42
n=24
n=29
n=859
n=549
n=176
n=13
0
20
40
60
80
100
120
phenotypically susceptible (%)
E
Discordance follows substrate preference
OXA-48 hydrolyses imipenem ~28x faster than meropenem (kcat 2.8 vs ~0.1 s⁻¹)
meropenem
imipenem
phenotypically susceptible (%)
216
29
1,415
1,047
221
32
OXA-48-like
KPC
NDM
F
MIC distribution by genotype
log2 meropenem MIC (mg/L)
S breakpoint
OXA-48-like (intact)
OXA-48-like (OmpK36 loop-3 insertion)
KPC (intact)
KPC (OmpK36 loop-3 insertion)
NDM (intact)
NDM (OmpK36 loop-3 insertion)